Ex-2

$$2x + 4y + 6z = 0$$
$$4x + 5y + 6z = 3$$
$$7x + 8y + 9z = 6$$

The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & 4 & 6 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 6 \end{array}\right] \underset{\sim}{\tfrac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 6 \end{array}\right] \quad \begin{array}{l} R_2 = R_2 - 4R_1 \\ R_3 = R_3 - 7R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 3 \\ 0 & -6 & -12 & 6 \end{array}\right] \underset{-\frac{1}{6}R_3}{\overset{-\frac{1}{3}R_2}{\sim}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -1 \end{array}\right] \quad \underset{\sim}{R_3 = R_3 - R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

leading entries
$\therefore$ (1,1) and (1,2) are pivot position
$\Rightarrow$ 1st & 2nd ~~cto~~ columns are pivot columns

The associated equations are

$$x + 2y + 3z = 0$$
$$y + 2z = -1$$
$$0 = 0$$

Here x and y are ~~pivot~~ basic variables.
$\therefore$ z is free variable.
$\therefore$ ~~If we~~ x and y are determined by z.

If we choose $z = t$, an arbitary real number, we get

$y = (-2t-1)$ and $x + 2(-2t-1) + 3t = 0 \Rightarrow x = t+2$

$\therefore (x, y, z) = (t+2, -2t-1, t) = (2, -1, 0) + t(1, -2, 1)$ $\rightarrow$ Equation of a st. line.

This line is the intersection of the planes $x+2y+3z=0$ and $y+2z=-1$

Thus the system has infinitely many solutions.

Ex-3:

$$x + 2y + 3z = 0$$
$$4x + 5y + 6z = 3$$
$$7x + 8y + 9z = 0$$

The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 0 \end{array}\right] \underset{R_3 = R_3 - 7R_1}{\overset{R_2 = R_2 - 4R_1}{\sim}} \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 3 \\ 0 & -6 & -12 & 0 \end{bmatrix} \overset{R_3 = R_3 - 2R_2}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 3 \\ 0 & 0 & 0 & -6 \end{array}\right]$$

The associated system of equations is

$$\left.\begin{aligned} x + 2y + 3z &= 0 \\ -3y + -6z &= 3 \\ 0 &= -6 \end{aligned}\right\}$$

Whatever values we choose for x, y, z the equation $0 = -6$ cannot be satisfied. Thus the system is inconsistent, i.e., it has no solution.

Concluding remarks: —

If REF of the coefficient matrix has no. of non-zero rows < no. of variables in the system $\Rightarrow$ The system cannot have exact one solution

If the REF of the augmented matrix has a row $[0 \cdots 0 | c]$, $c \neq 0$, then then system has no solution.

If the REF of the augmented matrix does not have a row $[0 \cdots 0 | c]$, $c \neq 0$, then the system has infinitely many solution.

System of ~~homog~~ Homogeneous Equations

If all the equations have constant term zero.

Ex:
$$2x + 3y = 0$$
$$5x - 2y = 0$$

Note: (1) $(0, 0, \ldots, 0)$ is always a solution of this system. So these systems have either an infinitely many solutions or one single solution $(0, \ldots, 0)$.

(2) For these system the augmented matrices have the last column zero. Thus the REF of any augmented matrix does not have a row $[0 \cdots 0 | c]$, $c \neq 0$

(3) Here, if REF of the coefficient matrix has no. of non-zero rows = no. of variables in the system
$\Rightarrow$ the system has exact one solution $(0, \ldots, 0)$, otherwise the system has infinitely many solutions.

Method 2: Gauss - Jordan Elimination Method

German engineer Wilhelm Jordan (1844–1899) popularized the (modified) Gauss elimination method, named as Gauss-Jordan elimination method. In ~~principle~~ principle this method well suited for solving linear systems on ~~to~~ a computer.

Goal: Transform the augmented matrix into Reduced-Row-Echelon Form (RREF) (by using elementary row operation

Reduced-Row-Echelon Form (RREF): A matrix is in RREF if

(a) it is in REF

(b) The leading entry in each row is 1.

(c) Each leading 1 is the only non zero entry in its column

Ex -
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 9 \\ 0 & 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Th:

~~Note~~: Gauss-Jordan elimination method produces a unique augmented ~~no~~ matrix in RREF.

Ex. of Gauss-Jordan elimination method

Ex-1

$$3x + 2y + z = 10$$
$$2x + 4y + 6z = 28$$
$$x + 3y + 2z = 13$$

The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 2 & 4 & 6 & 28 \\ 1 & 3 & 2 & 13 \end{array}\right] \xrightarrow[R_1 \leftrightarrow R_3]{\cancel{\text{by using}}\ \frac{1}{2}R} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 13 \\ 2 & 4 & 6 & 28 \\ 3 & 2 & 1 & 10 \end{array}\right] \quad \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & 13 \\ 0 & -2 & 2 & 2 \\ 0 & -7 & -5 & -29 \end{array}\right] \underset{-1R_3}{\overset{-\frac{1}{2}R_2}{\sim}} \left[\begin{array}{ccc|c} 1 & 3 & 2 & 13 \\ 0 & 1 & -1 & -1 \\ 0 & 7 & 5 & 29 \end{array}\right] \quad \begin{array}{l} R_1 = R_1 - 3R_2 \\ R_3 = R_3 - 7R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 16 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 12 & 36 \end{array}\right] \underset{\frac{1}{12}R_3}{\sim} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 16 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right] \quad \begin{array}{l} R_1 = R_1 - 5R_3 \\ R_2 = R_2 + R_3 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Now we are in RREF.

The associated equations are

$$x = 1$$
$$y = 2$$
$$z = 3$$

Ex-2

$$2x+4y+6z=0$$
$$4x+5y+6z=3$$
$$7x+8y+9z=6$$

The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & 4 & 6 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 6 \end{array}\right] \underset{\sim}{\tfrac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 6 \end{array}\right] \quad \begin{array}{l} R_2 = R_2 - 4R_1 \\ R_3 = R_3 - 7R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 3 \\ 0 & -6 & -12 & 6 \end{array}\right] \underset{\sim}{-\tfrac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & -6 & -12 & 6 \end{array}\right] \quad \begin{array}{l} R_1 = R_1 - 2R_2 \\ R_3 = R_3 + 6R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$ The associated equations are

$$\left.\begin{array}{r} x - z = 2 \\ y + 2z = -1 \\ 0 = 0 \end{array}\right\}$$

Here z is a free variable.
Let $z = t$, an arbitary real
$\therefore y = -2t - 1$
and $x = t + 2$

$\therefore$ The solutions are $(t+2, -2t-1, t)$ $\forall t \in \mathbb{R}$

$\therefore$ The system has infinitely many solutions.

Ex-3

$$x + 2y + 3z = 0$$
$$4x + 5y + 6z = 3$$
$$7x + 8y + 9z = 0$$

The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 3 \\ 7 & 8 & 9 & 0 \end{array}\right] \quad \begin{array}{l} R_2 = R_2 - 4R_1 \\ R_3 = R_3 - 7R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 3 \\ 0 & -6 & -12 & 0 \end{array}\right] \quad -\tfrac{1}{3}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & -6 & -12 & 0 \end{array}\right] \quad \begin{array}{l} R_1 = R_1 - 2R_2 \\ R_3 = R_3 + 6R_2 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & -6 \end{array}\right] \quad -\tfrac{1}{6}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} R_1 = R_1 - 2R_3 \\ R_2 = R_2 + R_3 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Now we are in RREF.

Here Gauss-Jordan elimination produced an augmented matrix row containing all zero coefficient ~~entry~~ entries with a non-zero constant entry i.e. $[0\ 0\ 0 \mid 1]$. So the system is inconsistant, i.e., ~~to~~ it has no solution

Minor of a Matrix

Let A be an $m \times n$ matrix and k be an integer with $0 < k \leq m$, and $0 < k \leq n$. The determinant of a $k \times k$ ~~squar~~ matrix obtained from A by deleting $m-k$ rows and $n-k$ ~~colo~~ columns, is called a minor of order k of A.

Ex- (1) $\begin{vmatrix} 1 & 4 \\ 9 & 12 \end{vmatrix}$ is a minor of order 2 of $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$

(2) $\begin{vmatrix} 1 & 3 & 4 \\ 5 & 7 & 8 \\ 9 & 11 & 12 \end{vmatrix}$ is a minor of order 3 of $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$

NB: A $m \times n$ matrix has $\binom{m}{m-k} \times \binom{n}{n-k}$ number of minors of order k.

Rank of a Matrix

The rank of a matrix $A_{m \times n}$ is the greatest positive integer r s.t. A has at least one non-zero minor of order r.

Ex- (1) Let $A = \begin{bmatrix} 2 & 3 & -1 & 1 \\ 3 & 0 & 4 & 2 \\ 6 & 9 & -3 & 3 \end{bmatrix}$. Here $R_3 = 3R_1$. Hence every minor of order 3 is zero. So rank of $A < 3$. Since there is a 2nd order minor $\begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} \neq 0 \Rightarrow \text{rank}(A) = 2$.

Ex-2: $A = \begin{bmatrix} 2 & 3 & 0 & 4 & -1 \\ 1 & 2 & 5 & -7 & 3 \\ 4 & 6 & 0 & 8 & -2 \\ 3 & 5 & 5 & -3 & 2 \end{bmatrix}$

Sol^n: Check all five 4-order minors of A.

→ You will get all of them are zero.

Now check all thirty minors of order 3.

→ All of them are zero

Next examine minors of order 2.

→ $\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \neq 0 \Rightarrow \text{rank}(A) = 2$

N.B.: This method of computing the rank is very laborious. ~~The~~ So we must look for a different method.

Good News ⇒ Th: The rank of a matrix remains invariant under an elementary row operation.

→ Th: If REF(A) has r nonzero rows, then

$$\text{rank}(A) = r.$$

Ex- $\begin{bmatrix} 2 & 3 & 0 & 4 & -1 \\ 1 & 2 & 5 & -7 & 3 \\ 4 & 6 & 0 & 8 & -2 \\ 3 & 5 & 5 & -3 & 2 \end{bmatrix} \underset{\sim}{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 5 & -7 & 3 \\ 2 & 3 & 0 & 4 & -1 \\ 4 & 6 & 0 & 8 & -2 \\ 3 & 5 & 5 & -3 & 2 \end{bmatrix}$

$R_2 = R_2 - 2R_1$
$R_3 = R_3 - 4R_1$
$R_4 = R_4 - 3R_1$

$$\begin{bmatrix} 1 & 2 & 5 & -7 & 3 \\ 0 & -1 & -10 & 18 & -7 \\ 0 & -2 & -20 & 36 & -14 \\ 0 & -1 & -10 & 18 & -7 \end{bmatrix}$$

$R_3 = R_3 - 2R_2$
$R_4 = R_4 - R_2$

$$\begin{bmatrix} 1 & 2 & 5 & -7 & 3 \\ 0 & -1 & -10 & 18 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This in REF. There are two non-zero rows.
Hence $\text{rank}(A) = 2$

<u>Note</u>: ① If $\text{rank}(A_{n\times n}) = n$, $n>0$, $\Longleftrightarrow$ $|A| \neq 0$.

$\therefore$ $\text{rank}(A_{m\times n}) < n \iff |A| = 0$

② $\text{rank}(\text{Augmented matrix } \cancel{\text{of } A}) \geq \text{rank}(\cancel{A})$ (coefficient matrix)

For a system of linear equations the rank of the augmented matrix is greater than equal to the rank of the coefficient matrix.

<u>Observation</u>: If $A_{m\times n}$ is a coefficient matrix then

no. of free variables + $\text{rank}(A_{m\times n}) = n$

Rank and the solutions of ~~the~~ system of linear equations

$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array}\right\} \Rightarrow AX = B$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- The system has m equations and n variables.
- For homogeneous system $b_i = 0 \ \forall i \Rightarrow B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0$ | If $B \neq \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$, the system is called non-homogeneous.

Let $\bar{A}$ be the augmented matrix.

Cases: $(m=n)$, $(m>n)$, $(m<n)$

Case (a): $\text{rank}(A) < \text{rank}(\bar{A})$

The system is inconsistent i.e. has no solution.

Case (c): $\text{rank}(A) = \text{rank}(\bar{A}) = n$ (Here no. of free variable $= 0$)

The system has the unique solution $X = A^{-1}B$

NB: (i) When $m<n$, this case will not arise $\because \text{rank}(\bar{A}) \leq m < n$.

Case (b): $\text{rank}(A) = \text{rank}(\bar{A}) < n$ (ii) When $m=n$, $\text{rank}(A) = n \Leftrightarrow$ unique sol^n

The system has infinitely many solutions.

(Here no. of free variables > 0)

Th: ~~The~~ A non-homogeneous system $AX = B$ is consistent iff $\text{rank}(A) = \text{rank}(\bar{A})$

NB: For a homogeneous system $AX = 0$, $\text{rank}(A) = \text{rank}(\bar{A})$. Thus it always has a solution.